Goutam Sachdev

AI ENGINEER

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Summary

AI Software Engineer with 2 years of experience building production-ready AI solutions, including LLM-powered chatbots, agentic AI automation, and computer vision applications. Skilled in Python, modern ML frameworks, backend engineering, and frontend development using React and Next.js. Delivered scalable systems that improved customer support efficiency, streamlined workflows, and reduced operational costs.

Technical Skills

Programming: Python, C++, JavaScript, SQL

Frameworks & Libraries: FastAPI, Django, Flask, ReactJS, OpenCV, PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, LangChain, CrewAI, LangGraph

Databases & Storage: MySQL, PostgreSQL, MongoDB, Redis, ChromaDB, Pinecone

AI & Agentic Systems: Agentic AI, AI Agents, LLM Agent Frameworks, Multi-Agent Systems, Prompt Engineering, Retrieval-Augmented Generation (RAG), Graph RAG, Generative AI

Core Skills: REST API Development, Backend Development, API Integration, Computer Vision

Tools & Platforms: Git, Docker, LiveKit, n8n

Professional Experience

Techno Channels Karachi, PK

AI Engineer -Hybrid *April 2026 – Present*

- Own the backend of **SHIFU**, a multi-tenant AI assistant platform in **Python** and **FastAPI** serving **multiple customer workspaces**, covering workspaces, bots, source ingestion, and bot deployment.
- Shipped **the RAG pipeline** end to end - chunking and embedding customer documents into **PostgreSQL/pgvector**, retrieving over the customer's own corpus, and generating grounded answers with source citations.
- Designed the model layer: a logical model catalog mapped to provider deployments behind an internal gateway, with tier-based routing driven by plan entitlements and per-plan credit metering via atomic reserve-before-work deduction, reducing inference costs by serving eligible requests through lower-cost models.
- Implemented tenant isolation, role-based access control, and an append-only **audit log** recording actor, target, and timestamp for every workspace action, providing complete workspace activity tracking and traceability.
- Automated document ingestion and bot retraining as a background pipeline on **Celery** and **Redis**, enabling uploaded content to become searchable without blocking live query traffic.

Quentis Karachi, PK

AI Python Developer *May 2025 – April 2026*

- Developed the real-time conversational frontend using **Next.js** and the **LiveKit React framework**, implementing microphone access, real-time audio streaming, participant state handling, and responsive UI components for live AI conversations.
- Built a modern chat interface using **Next.js, React Hooks, and Tailwind CSS**, ensuring real-time message updates, session persistence, and smooth user experience across voice and text-based interactions.
- Implemented an AI-driven appointment scheduling system within chatbot workflows, allowing users to book and manage appointments while preserving conversational context.
- Built a customer support chatbot using **OpenAI SDK with ChatKit**, implementing a **RAG-based architecture** and integrated appointment scheduling, delivering consistent responses with average latency under **2 seconds** and enabling seamless **AI-to-human escalation** flows.
- Implemented the chatbot frontend using **Next.js and OpenAI ChatKit**, handling real-time message streaming, conversation state management, optimistic UI updates, and reusable chat components for scalable deployment.
- Integrated **Zendesk (CRM)** for ticket creation, updates, and agent assignment, ensuring smooth handoff between AI and human agents while preserving conversation context and minimizing ambiguity.

- Optimized backend services, frontend state management, and system performance to maintain **low latency, high reliability**, and stable conversational flows across both fully automated and **hybrid (AI + human)** customer support workflows.
- Worked on a **washery mobile application** managing a scalable Django backend, implementing customer and driver modules, JWT-based authentication, RESTful APIs, and optimized order lifecycle workflows for cloth pickup and delivery; ensured **low-latency API responses (¡200ms)** and efficient database operations using PostgreSQL.
- Worked on a **hybrid RAG system** using LlamaIndex, combining **Graph RAG** and **vector retrieval (Deep Lake)** for improved multi-hop reasoning; built a document processing pipeline using Docling for structured context extraction (text, tables, charts), applied advanced chunking and embedding strategies, and integrated **NVIDIA Nemotron Nano VLM** for multimodal understanding of graph images, achieving **sub-2s response latency** and significantly improving retrieval accuracy and answer relevance.

BitsWits Karachi, PK

AI Python Developer Internship *Feb 2025 – May 2025*

- Built and deployed AI-powered chatbots using Django, LangChain, CrewAI, and LangGraph, reducing manual customer support workload by **35%** and handling **500+ daily user queries** with <2s latency.
- Developed a generative AI assistant for Shopify and WordPress e-commerce stores using **LangGraph + FastAPI**, enabling natural language product search, cart management, and **in-chat product image rendering with AI-generated product descriptions** through an embedded chatbot interface; improved product discovery speed by **80%** while maintaining real-time inventory synchronization via PostgreSQL.
- Built a real-time customer support chatbot using **LiveKit (WebRTC)** with a **RAG-based backend** powered by Sentence Transformers, LlamaIndex, and ChromaDB; designed FastAPI services for efficient knowledge retrieval and low-latency responses.

Projects

AI Voice Agent System (Restaurant Reservation) — Built a real-time voice AI agent that can **call and converse with humans** to handle restaurant reservations end-to-end, using a dual-agent architecture with **LiveKit, Twilio, Groq, and Cartesia TTS**; implemented **Whisper-based STT, VAD, and interruption handling** for natural, low-latency interactions.

Schedule Assistant — Automated scheduling tool integrating Groq API, Gmail API, and Google Calendar. Used LangGraph to resolve conflicts and cut scheduling time by **60%**.

Goti: Advanced Voice Assistant Real-time assistant leveraging LLaMA 3 + custom deep-learning transcription.

Pediatric Wrist Report Generation App, SSD Hospital Sukkur — Developed an AI-powered application that auto-generates 90% of pediatric wrist radiology reports. **Used generative models trained on 1200+ custom hospital reports to achieve automation. Integrated a frontend interface for radiologists with dynamic report population and editing.**

AutoBlackjack Computer vision assistant using YOLOv8 + OpenCV + Hi-Lo algorithm. Provided real-time card detection with **95% detection accuracy** and live betting recommendations (educational use).

Education

Sukkur IBA University Sukkur, PK

Bachelor of Computer Science CGPA 3.01/4.00 *Sep 2020 – Aug 2024*

- Relevant Coursework: Artificial Intelligence, Computer Vision, Web Engineering, Data Structures

Certificates

Claude 101 & **Claude Code in Action** *Anthropic*

AI Agents Fundamentals & Hugging Face Agents Course *Hugging Face*

Building AI Voice Agents for Production *DeepLearning.AI*

Practical Multi AI Agents and Advanced Use Cases with CrewAI *DeepLearning.AI*

Intermediate Python *DataCamp*

Problem Solving Certificate *HackerRank*

Trek Venture 6.0 – Leadership & Experiential Learning Program *Million Smiles*